- Cake Production System
  - BAT-FEAT-00 Batter Preparation
    - BAT-FEAT-01 Ingredient Intake & Dosing
      - BAT-FEAT-02 Order and ingredient intake
      - BAT-FEAT-03 Recipe-target dispensing
      - BAT-FEAT-04 Dose weighing and verification
      - BAT-FEAT-05 Top-up on under-dose
      - BAT-FEAT-06 Over-dose batch discard
      - BAT-FEAT-07 Repeated-failure lockout
    - BAT-FEAT-08 Mixing
      - BAT-FEAT-09 Recipe-timed mixing
      - BAT-FEAT-10 Consistency monitoring
      - BAT-FEAT-11 Batch release gate
        - BAT-FEAT-12 Human inspection
      - BAT-FEAT-14 Re-mix of a rejected batch
      - BAT-FEAT-15 Mixer fault handling
    - BAT-FEAT-16 Batter Handoff
      - BAT-FEAT-17 Batter-ready handoff
      - BAT-FEAT-18 Batter waiting-time limit
      - BAT-FEAT-19 Hold on depositing stopped
    - BAT-FEAT-20 Line health and reporting
      - BAT-FEAT-21 Event reporting
      - BAT-FEAT-22 Health reporting
  - THM-FEAT-00 Thermal Processing
    - THM-FEAT-01 Depositing
      - THM-FEAT-02 Pan-presence check
      - THM-FEAT-03 Fill-weight check
      - THM-FEAT-04 Repeated-failure lockout
      - THM-FEAT-05 New-pan admission stop
      - THM-FEAT-06 Reduced filling on low pan stock
    - THM-FEAT-07 Baking
      - THM-FEAT-08 Zone temperature control
      - THM-FEAT-09 Zone-fault escalation
    - THM-FEAT-10 Tunnel-Exit Inspection
      - THM-FEAT-11 Exit doneness check
      - THM-FEAT-12 Refuse diversion and discard logging
      - THM-FEAT-13 Camera-failure fallback
    - THM-FEAT-14 Cooling
      - THM-FEAT-15 Cooling control
      - THM-FEAT-16 Cooling-failure response
        - THM-FEAT-17 Cooling buffer fallback
          - THM-FEAT-18 Buffer hold and release
          - THM-FEAT-19 Buffer wait-time limit
          - THM-FEAT-20 Buffer-full handling
        - THM-FEAT-21 Discard fallback
    - THM-FEAT-22 Line health and coordination
      - THM-FEAT-23 Health reporting
      - THM-FEAT-24 Band-state response
      - THM-FEAT-25 Event reporting
        - THM-FEAT-27 Log events
        - THM-FEAT-28 Warning events
        - THM-FEAT-29 Error events
      - THM-FEAT-26 Recovery coordination
  - FIN-FEAT-00 Finishing & Packaging
    - FIN-FEAT-01 Depanning
      - FIN-FEAT-02 Stuck-cake rejection
      - FIN-FEAT-03 Hold and recover on depanning faults
    - FIN-FEAT-04 Quality Inspection
      - FIN-FEAT-05 Weight check
      - FIN-FEAT-06 Visual check
      - FIN-FEAT-07 Metal check
      - FIN-FEAT-08 Rejection handling and reporting
    - FIN-FEAT-09 Packaging
      - FIN-FEAT-10 Wrapping
      - FIN-FEAT-11 Cartoning
      - FIN-FEAT-12 Case-weight check
      - FIN-FEAT-13 Packaging reject handling
      - FIN-FEAT-14 Holding area with stop/resume
    - FIN-FEAT-15 Palletising
    - FIN-FEAT-16 Pan Loop
      - FIN-FEAT-18 Pan stock management
      - FIN-FEAT-19 Washing catch-up after a fault
    - FIN-FEAT-20 Line health and coordination
      - FIN-FEAT-21 Health reporting
      - FIN-FEAT-22 Band-state response
  - MON-FEAT-00 Line Monitoring & Health + Band Transport
    - MON-FEAT-01 Health Monitoring
    - MON-FEAT-02 Traceability & Event Logging
    - MON-FEAT-03 Maintenance Management
    - MON-FEAT-04 Band Coordination
      - MON-FEAT-05 Band Speed Supervision
      - MON-FEAT-06 Pan Position & Spacing Tracking
    - MON-FEAT-07 Container Recovery
  - PLM-FEAT-00 Production & Logistics Management
    - PLM-FEAT-01 Order and Scheduling
      - PLM-FEAT-02 Production order creation and holding
      - PLM-FEAT-03 Warehouse timeout handling
    - PLM-FEAT-04 Recipe Management
      - PLM-FEAT-05 Recipe target definition
    - PLM-FEAT-06 Warehouse and Inventory
      - PLM-FEAT-07 Ingredient reservation processing
      - PLM-FEAT-08 Organic ingredient verification
      - PLM-FEAT-09 Inventory level updating
      - PLM-FEAT-10 Insufficient stock handling and reporting
    - PLM-FEAT-11 Transport and Shipment
      - PLM-FEAT-12 Pallet receiving and sending coordination

● Mandatory ○ Optional) Alternative (XOR)) Or group